

Auxiliary Gradient-Flow Solvers for Generalized Newtonian Models

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Overview

- 1 Introduction
- 2 Continuous theory
- 3 FEM solver
- 4 Numerical results
- 5 Future work

Preprint: <https://arxiv.org/abs/2606.05926>

Implementation:

https://github.com/stefanozampini/generalized_newtonian

Model energy

Interested in the numerical solution of minimization problems of the type

$$\operatorname{argmin}_{u \in V} E(u), \quad E(u) := \int_{\Omega} \phi(|Du|) \, dx - F(u), \quad (1)$$

where $\Omega \subset \mathbb{R}^d$, $d = 2, 3$, ϕ is an N-function¹,

$$D = \nabla, \quad \text{or} \quad D = \varepsilon(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T)$$

depending on whether u is a scalar or an incompressible vector field in an Orlicz-Sobolev functional space

$$V = W_0^{1,\phi}(\Omega), \quad \text{or} \quad V = \left\{ \mathbf{v} \in W_0^{1,\phi}(\Omega; \mathbb{R}^d) : \nabla \cdot \mathbf{v} = 0 \right\}.$$

The associated Euler–Lagrange equations are nonlinear and may be singular or degenerate, which present challenges for both analysis and numerical approximation.

¹Musielak 1983.

FEM literature

- Classical works on power-law and Carreau models²
- FEM optimality (in terms of quasi-norms)³
- Adaptive finite element methods for the p -Laplacian⁴
- FEM for p -Stokes⁵
- High-order FEM approximation properties⁶
- DG for quasilinear elliptic problems⁷
- Hybrid High-Order methods⁸
- Virtual Element methods⁹

²Glowinski and Marroco 1975; Wei 1989; Barrett and Wen Bin Liu 1993; Barrett and W. Liu 1993; Barrett and W. Liu 1994.

³Ebmeyer and W. Liu 2005.

⁴Diening and Kreuzer 2008; Belenki, Diening, and Kreuzer 2012; Carstensen and Klose 2003.

⁵Belenki, Berselli, et al. 2012.

⁶Ainsworth and Kay 2000.

⁷Houston, Robson, and Süli 2005; Houston, Süli, and Thomas P. Wihler 2008.

⁸Botti et al. 2021; Di Pietro and Droniou 2017; Carstensen and Tran 2021.

⁹Antonietti, Veiga, Botti, Vacca, et al. 2024; Antonietti, Veiga, Botti, Harnist, et al. 2026.

Kačanov method

Newton's method can break down. Today's gold-standard alternative is the Kačanov method

- Originated in nonlinear magnetostatics¹⁰
- Later developed as a general fixed-point linearization technique for nonlinear PDEs and variational inequalities¹¹.
- Needs to be *relaxed* for some energies of the type (1): convergence analysis for
 - p -Laplacian¹²
 - shear-thinning Carreau-type fluids¹³
 - regularized Bingham models¹⁴.
- A damped version in¹⁵

¹⁰Kačur et al. 1968.

¹¹Fučík, Kratochvíl, and Nečas 1974; Han, Jensen, and Shimansky 1997; Nečas and Hlaváček 1983; Zeidler 2013.

¹²Diening, Fornasier, et al. 2020; Balci, Diening, and Storn 2023.

¹³Heid and Süli 2022b.

¹⁴Heid and Süli 2022a.

¹⁵Heid and Thomas P Wihler 2022.

From biological transportation networks to p -Laplacian

The starting point of the present work is the DAE¹⁶

$$\frac{ds}{dt} = |\nabla u|^2 - s^{\gamma-1}, \quad -\nabla \cdot ((s+r)\nabla u) = f, \quad \gamma > 1, r > 0,$$

The steady states solve the regularized p -Laplacian equations

$$-\nabla \cdot (|\nabla u|^{p-2} + r)\nabla u) = f, \quad p = \frac{2\gamma}{\gamma-1} > 2.$$

Rewriting γ in terms of p , set $c = s^{\frac{2}{p-2}}$, the dynamics become

$$\frac{dc}{dt} = \frac{2}{p-2} c^{\frac{4-p}{2}} (|\nabla u|^2 - c), \quad -\nabla \cdot \left(\left(c^{\frac{p-2}{2}} + r \right) \nabla u \right) = f.$$

If the goal is to recover only the steady state, one is thus tempted to use

$$\frac{dc}{dt} = |\nabla u|^2 - c, \quad -\nabla \cdot \left(c^{\frac{p-2}{2}} \nabla u \right) = f.$$

¹⁶Haskovec, Markowich, and Zampini 2026.

Comparison vs Kačanov

The Kačanov iteration for the p -Laplacian

$$\int_{\Omega} |\nabla u_n|^{p-2} \nabla u_{n+1} \cdot \nabla v \, dx = F(v).$$

The relaxed Kačanov iteration (converges for $p < 2$ only¹⁷)

$$\int_{\Omega} \max\{\min\{|\nabla u_n|^{p-2}, \varepsilon_+\}, \varepsilon_-\} \nabla u_{n+1} \cdot \nabla v \, dx = F(v).$$

This work

$$\begin{aligned} \int_{\Omega} \frac{c_{n+1} - c_n}{\tau_n} w &= \int_{\Omega} (|\nabla u_{n+1}|^2 - c_{n+1}) w \, dx, \\ \int_{\Omega} c_{n+1}^{(p-2)/2} \nabla u_{n+1} \cdot \nabla v \, dx &= F(v). \end{aligned}$$

¹⁷Diening, Fornasier, et al. 2020.

Contributions

- Auxiliary energy with domain of arbitrarily truncated functions
- Continuous analysis using the theory of gradient flows in metric spaces¹⁸
- Exponential convergence of the method of minimizing movements¹⁹ toward the minimizers
- Rigorous convergence theory: e.g. for power-laws $4/3 \leq p \leq 4$, $p \neq 2$
- FEM solver building on the metric gradient flow theory²⁰; metric dependent Riesz-map allows evolutions of the type

$$\frac{dc_h}{dt} = |Du_h|^2 - c, \quad \int_{\Omega} \mu(c_h) Du_h : Dv_h \, dx = F(v_h), \quad \mu(r) := \frac{\phi'(\sqrt{r})}{\sqrt{r}}.$$

- DAE has global solutions, converging to the solution of

$$\int_{\Omega} \mu(|Du_h|^2) Du_h : Dv_h \, dx = F(v_h)$$

¹⁸Ambrosio, Gigli, and Savaré 2005.

¹⁹De Giorgi 1993.

²⁰Liero and Mielke 2013.

Functional spaces

Analysis in Orlicz-Sobolev spaces $W^{1,\phi} = \{v \in L^\phi(\Omega) : \nabla v \in L^\phi(\Omega; \mathbb{R}^d)\}$
where the Orlicz space is

$$L^\phi(\Omega) = \left\{ v \in L^1_{\text{loc}} : \int_{\Omega} \phi(|v|) \, dx < +\infty \right\}$$

equipped with the Luxemburg norm

$$\|v\|_{L^\phi(\Omega)} := \inf \left\{ \lambda > 0 : \int_{\Omega} \phi(|v|/\lambda) \, dx \leq 1 \right\}.$$

Example: power-law model

$$\phi(t) = \frac{t^p}{p}, \quad L^\phi(\Omega) = L^p(\Omega), \quad p > 1.$$

Some assumptions

Separable and reflexive Banach space under

Assumption

The Simonenko indices of ϕ^a are bounded, i.e.

$$R_{\phi}^{-} := \inf_{r>0} \frac{r\phi'(r)}{\phi(r)} > 1, \quad R_{\phi}^{+} := \sup_{r>0} \frac{r\phi'(r)}{\phi(r)} < +\infty.$$

^aSimonenko 1963.

Nice regularity theory under the assumption

Assumption

ϕ is uniformly convex^a

$$S_{\phi}^{-} := \inf_{r>0} \frac{r\phi''(r)}{\phi'(r)} > 1, \quad S_{\phi}^{+} := \sup_{r>0} \frac{r\phi''(r)}{\phi'(r)} < +\infty.$$

^aDiening and Kreuzer 2008.

Metric space

We use

$$\Phi(r) := \phi(\sqrt{r}), \quad \Phi(r) = \frac{1}{2} \int_0^r \mu(s) \, ds, \quad \mu(r) := 2\Phi'(r) = \frac{\phi'(\sqrt{r})}{\sqrt{r}}.$$

$L^\Phi(\Omega)$ is not always an Orlicz space. New theory is based on

Assumption

The function Φ is either strictly convex (i.e. $\mu'(r) > 0$) or strictly concave ($\mu'(r) < 0$) on $(0, +\infty)$.

$L^\Phi(\Omega)$ is a complete geodesic metric space (isometric to $L^2(\Omega)$) with

$$d(c_1, c_2) = \|\psi(c_1) - \psi(c_2)\|_{L^2(\Omega)},$$

$$\psi(r) := \operatorname{sgn}(r) \int_0^{|r|} \sqrt{\frac{\omega}{2} \mu'(s)} \, ds, \quad \omega := \begin{cases} 1, & \Phi \text{ convex,} \\ -1, & \Phi \text{ concave,} \end{cases}$$

Auxiliary energy

Using a tangent approximation of Φ , we can define the auxiliary energy $\mathcal{E}_\epsilon : L_\epsilon^\Phi(\Omega) \rightarrow (-\infty, +\infty]$ such that

$$\mathcal{E}_\epsilon(c) := -\omega \left(\int_\Omega \Phi(c) - \frac{1}{2}\mu(c)c + \frac{1}{2}|Du[c]|^2 dx - F(u[c]) \right),$$

$$u[c] := \operatorname{argmin}_{u \in W} \left(\int_\Omega \frac{1}{2}\mu(c)|Du[c]|^2 dx - F(u) \right).$$

where, for an arbitrary $\epsilon \in (0, 1)$

$$L_\epsilon^\Phi(\Omega) := \left\{ c \in L^\Phi(\Omega) : \epsilon \leq c(x) \leq 1/\epsilon \text{ a.e. in } \Omega \right\},$$

$$W = H_0^1(\Omega), \quad \text{or} \quad W = \left\{ \mathbf{v} \in H_0^1(\Omega; \mathbb{R}^d) : \nabla \cdot \mathbf{v} = 0 \right\}.$$

$\mathcal{E}_\epsilon$ is lower-semicontinuous and bounded from below

$$-\omega E(u[c]) \leq \mathcal{E}_\epsilon(c), \quad c \in L_\epsilon^\Phi(\Omega).$$

Gradient flow in metric spaces

Existence and uniqueness of the minimizers are proven using the theory of gradient flows in metric spaces²¹. Evolution defined by the *method of minimizing movements*²²

$$c_{k+1} \in \min_{c \in L^{\Phi}(\Omega)} \mathcal{E}_{\epsilon}(c) + \frac{1}{2\tau} d(c_k, c)^2,$$

We have the a-priori estimates

$$d(\bar{c}_{\tau}(t), c^*) \leq C e^{-\lambda_{\tau} t}, \quad \lambda_{\tau} := \frac{\log(1 + \lambda\tau)}{\tau}. \quad (2)$$

$\bar{c}_{\tau}(t)$ the left-continuous piecewise constant interpolant of the minimizing movement iteration, c^* the minimizer of the energy, λ the geodesical convexity constant, i.e.

$$\mathcal{E}_{\epsilon}(\gamma_g(s)) \leq (1-s)\mathcal{E}_{\epsilon}(c_0) + s\mathcal{E}_{\epsilon}(c_1) - \frac{\lambda}{2} s(1-s) d(c_0, c_1)^2, \quad \forall s \in [0, 1].$$

²¹Ambrosio, Gigli, and Savaré 2005.

²²De Giorgi 1993.

Convergence rate theory

Theorem

Assume $\inf_{r>0} \left\{ 1 + \frac{r\mu''(r)}{2\mu'(r)} \right\} > -\infty$. Then

(i) if Φ is convex and $\mu''(r) \leq 0$, or

(ii) if Φ is concave and $\mu''(r) \geq \frac{4\mu'(r)^2}{\mu(r)}$,

$$\lambda = \inf_{r>0} \left\{ 1 + \frac{r\mu''(r)}{2\mu'(r)} \right\}.$$

Remark

For the power-law case, a straightforward calculation shows

$$\lambda = \frac{p}{4} > 0, \quad 4/3 \leq p \leq 4, p \neq 2.$$

FEM approximation

The primal problem leads to the Euler–Lagrange equations

$$\int_{\Omega} A(D\bar{u}_h) : Dv_h \, dx = F(v_h), \quad \forall v_h \in W_h. \quad (3)$$

Optimal error²³ in terms of

$$\|V(D\bar{u}) - V(D\bar{u}_h)\|_{L^2(\Omega)},$$

where $\bar{u}$ is the minimizer of problem (1), and

$$A(P) := \mu(|P|^2)P = \frac{\phi'(|P|)}{|P|}P, \quad V(P) := \sqrt{\mu(|P|^2)}P, \quad P \in \mathbb{R}^{Nd}.$$

²³Ebmeyer and W. Liu 2005; Diening and Kreuzer 2008.

FEM discretization

$C_h \subset L^\Phi(\Omega)$, $U_h \subset H_0^1(\Omega)$, $\mathbf{U}_h \subset H_0^1(\Omega; \mathbb{R}^d)$, and $Q_h \subset Q$ respectively, and assume

Assumption

$C_h \subset L^\infty(\Omega)$, $|\nabla u_h|^2 \in C_h$ for all $u_h \in U_h$, $|\varepsilon(\mathbf{u}_h)|^2 \in C_h$ for all $\mathbf{u}_h \in \mathbf{U}_h$, and $\mathbf{U}_h \times Q_h$ is an inf-sup stable finite element pair for the weighted Stokes system.

For simplices

$$U_h^{k,T} := \{u_h \in H_0^1(\Omega) : u_h \in C^0(\Omega), u_h|_T \in \mathcal{P}_k(T)\},$$

$\nabla u_h|_T \in \mathcal{P}_{k-1}(T)$, thus

$$C_h^{k,T} := \{c_h \in L^\Phi(\Omega) : c_h|_T \in \mathcal{P}_{2k-2}(T)\}.$$

FEM discretization

$C_h \subset L^\Phi(\Omega)$, $U_h \subset H_0^1(\Omega)$, $\mathbf{U}_h \subset H_0^1(\Omega; \mathbb{R}^d)$, and $Q_h \subset Q$ respectively, and assume

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For tensor-product cells

$$U_h^{k,K} := \{u_h \in H_0^1(\Omega) : u_h \in C^0(\Omega), u_h|_K \in \mathcal{Q}_k(K)\},$$

$\nabla u_h|_K \in \mathcal{Q}_k(K)$, thus

$$C_h^{k,K} := \{c_h \in L^\Phi(\Omega) : c_h|_K \in \mathcal{Q}_{2k}(K)\}.$$

FEM discretization

$C_h \subset L^\Phi(\Omega)$, $U_h \subset H_0^1(\Omega)$, $\mathbf{U}_h \subset H_0^1(\Omega; \mathbb{R}^d)$, and $Q_h \subset Q$ respectively, and assume

Assumption

$C_h \subset L^\infty(\Omega)$, $|\nabla u_h|^2 \in C_h$ for all $u_h \in U_h$, $|\varepsilon(\mathbf{u}_h)|^2 \in C_h$ for all $\mathbf{u}_h \in \mathbf{U}_h$, and $\mathbf{U}_h \times Q_h$ is an inf-sup stable finite element pair for the weighted Stokes system.

Extends naturally to the vector-valued case. In this talk we use standard Taylor–Hood pairs to discretize the problem.

First variation of energy

We work with

$$C_h^+ := \left\{ c_h \in C_h : \operatorname{ess\,inf}_{x \in \Omega} c_h(x) > 0 \right\}.$$

First variation for the scalar problem

$$\mathcal{D}\mathcal{E}_h(c_h)[w_h] = \int_{\Omega} \frac{-\omega\mu'(c_h)}{2} (|\nabla u_h[c_h]|^2 - c_h) w_h \, dx.$$

First variation for the incompressible case

$$\mathcal{D}\mathcal{E}_h(c_h)[w_h] = \int_{\Omega} \frac{-\omega\mu'(c_h)}{2} (|\varepsilon(\mathbf{u}_h[c_h])|^2 - c_h) w_h \, dx.$$

Riesz map

$\mathcal{DE}_h(c_h) \in C_h^*$, turned into a gradient²⁴, by solving the variational problem

$$g_{c_h}(\nabla_{g_{c_h}} \mathcal{E}_h(c_h), w_h) = \mathcal{DE}_h(c_h)[w_h], \quad \forall w_h \in C_h,$$

where the inner product arising from the discrete metric tensor is

$$g_{c_h}(z_h, w_h) := (\mathcal{D}\psi(c_h)z_h, \mathcal{D}\psi(c_h)w_h)_{L^2(\Omega)} = \int_{\Omega} \frac{\omega}{2} \mu'(c_h) z_h w_h \, dx.$$

The Riesz map

$$\mathcal{G}(c_h) : C_h \rightarrow C_h^*, \quad \langle \mathcal{G}(c_h)w_h, z_h \rangle = g_{c_h}(z_h, w_h), \quad c_h \in C_h^+, \quad w_h, z_h \in C_h$$

is explicitly known, and we do not need to solve any additional variational problem to define the gradient

$$\nabla_{g_{c_h}} \mathcal{E}_h(c_h) = c_h - |Du_h[c_h]|^2.$$

²⁴Liero and Mielke 2013.

ODE

We can then consider the ODE

$$\begin{cases} \frac{dc_h}{dt} = |Du_h[c_h]|^2 - c_h, \\ c_h(0) = c_{h,0} \in C_h^+. \end{cases} \quad (4)$$

Theorem

The ODE problem (4) has a unique global solution $c_h \in C^1([0, +\infty); C_h)$ and $c_h(t) \in C_h^+$. Moreover, for each fixed time $T < +\infty$, there exists $\epsilon(T) > 0$ such that

$$\epsilon(T) \leq c_h(t) \leq 1/\epsilon(T), \quad \forall 0 \leq t \leq T.$$

ODE

We can then consider the ODE

$$\begin{cases} \frac{dc_h}{dt} = |Du_h[c_h]|^2 - c_h, \\ c_h(0) = c_{h,0} \in C_h^+. \end{cases} \quad (4)$$

Theorem

There exists a constant $C > 0$, independent of t , such that

$$\|V(Du_h[c_h(t)]) - V(D\bar{u}_h)\|_{L^2(\Omega)} \leq Ce^{-\lambda t} \sqrt{d_h(0)},$$

where $c_h(t)$ is the solution of the ODE (4), $\bar{u}_h$ is FEM solution of the primal problem (3) and λ is given as in the continuous case. Therefore, if $\lambda > 0$, we have

$$\lim_{t \rightarrow \infty} \|V(Du_h[c_h(t)]) - V(D\bar{u}_h)\|_{L^2(\Omega)} = 0.$$

Time advance: backward Euler

Given $(u_{h,n}, c_{h,n})$ and a current time step τ_n , find $(u_{h,n+1}, c_{h,n+1})$ such that for all $(v_h, w_h) \in U_h \times C_h$

$$\int_{\Omega} \frac{c_{h,n+1} - c_{h,n}}{\tau_n} w_h = \int_{\Omega} (|\nabla u_{h,n+1}|^2 - c_{h,n+1}) w_h \, dx,$$

$$\int_{\Omega} \mu(c_{h,n+1}) \nabla u_{n+1} \cdot \nabla v_h \, dx = F(v_h)$$

Preserves strict positivity of c_h between time steps, requires a Newton solve at each time step (does not preserve strict positivity between successive iterates).

Time advance: operator splitting

Given $(u_{h,n}, c_{h,n})$ and a current time step τ_n , find $(u_{h,n+1}, c_{h,n+1})$ such that for all $(v_h, w_h) \in U_h \times C_h$

$$\int_{\Omega} \frac{c_{h,n+1} - c_{h,n}}{\tau_n} w_h = \int_{\Omega} (|\nabla u_{h,n}|^2 - c_{h,n+1}) w_h \, dx,$$

$$\int_{\Omega} \mu(c_{h,n+1}) \nabla u_{h,n+1} \cdot \nabla v_h \, dx = F(v_h)$$

Preserves strict positivity of c_h between time steps, fully decouples the equations. First equation can be solved exactly by interpolation

$$c_{h,n+1} = \frac{\tau_n}{1 + \tau_n} |\nabla u_{h,n}|^2 + \frac{1}{1 + \tau_n} c_{h,n}.$$

Second equation requires only a linear solve per step.

Does not always lead to a convergent time iteration. Kačanov corresponds to $\tau_n = +\infty$.

Time advance: pseudo-time continuation like

Intermediate update $c_{h,n+1/2} = \frac{\tau_n}{1+\tau_n} |\nabla u_{h,n}|^2 + \frac{1}{1+\tau_n} c_{h,n}$, then solve

$$\begin{bmatrix} (\tau_n^{-1} + 1)M & -2B^n \\ C^n & K^n \end{bmatrix} \begin{bmatrix} \delta c_h \\ \delta u_h \end{bmatrix} = \begin{bmatrix} 0 \\ g \end{bmatrix},$$

where the matrix and vector entries are

$$M_{i,j} = \int_{\Omega} \phi_i \phi_j \, dx,$$

$$B_{i,j}^n = \int_{\Omega} \phi_i \nabla u_{h,n} \cdot \nabla \phi_j \, dx,$$

$$C_{i,j}^n = \int_{\Omega} \mu'(c_{h,n+1/2}) \phi_j \nabla u_{h,n} \cdot \nabla \phi_i \, dx,$$

$$K_{i,j}^n = \int_{\Omega} \mu(c_{h,n+1/2}) \nabla \phi_i \cdot \nabla \phi_j \, dx,$$

$$g_i = F(\phi_i) - \int_{\Omega} \mu(c_{h,n+1/2}) \nabla u_{h,n} \cdot \nabla \phi_i \, dx,$$

Requires two cheap linear solves on c_h and one Schur complement solve (symmetric positive definite)

$$S^n = K^n + \frac{2\tau_n}{1+\tau_n} C^n M^{-1} B^n.$$

Validation of λ -convexity estimates

Power-law, scalar case. *Bump* example from²⁵

$$u_e(x) = (x_1^2 - 1)(x_2^2 - 1), \quad \Omega = [0, 1]^2.$$

Using a 32×32 grid of quads, $k = 2$.

From (2) we expect convergence rates to be lower-bounded by

$$\lambda_\tau = \frac{\log(1 + \lambda\tau)}{\tau}, \quad \lambda = \begin{cases} p - 1, & p < 4/3, \\ \frac{p}{4}, & \frac{4}{3} \leq p \leq 4, p \neq 2, \\ 1, & p > 4. \end{cases}$$

²⁵Diening, Fornasier, et al. 2020, Section 6.

Validation of λ -convexity estimates

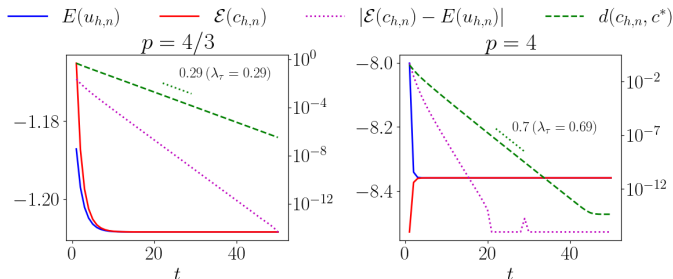


Figure: Representative convergence curves for the p -Laplacian problem using the BE method with $\tau = 1$ for different values of p as shown on top. Values for $E(u_{h,n})$ (blue continuous lines) and $\mathcal{E}(c_{h,n})$ (red continuous lines) are given on the left y -axis, while $d(c_{h,n}, c^*)$ (green dashed lines) and $|\mathcal{E}(c_{h,n}) - E(u_{h,n})|$ (magenta dotted lines) are given in logarithmic scale on the right axis. Experimental and theoretical convergence rates (in parentheses) are reported in the text.

Validation of λ -convexity estimates

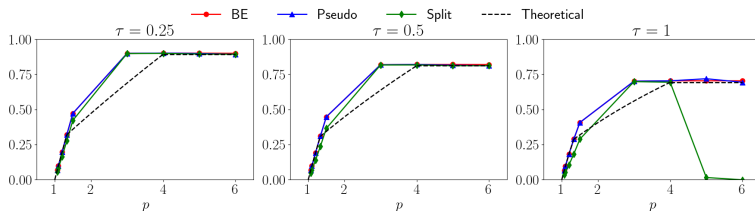


Figure: Comparison of experimental (continuous lines) and theoretical (dashed) convergence rates of BE (red), Pseudo (blue), and Split (green) methods for the p -Laplacian problem as a function of p for different values of τ as shown on top of each panel.

Validation of λ -convexity estimates

Incompressible vector valued case

$$\mathbf{u}_e(x) = 4(1 - x_1^2 - x_2^2)(x_2, -x_1), \quad p_e(x) = x_1^2 + x_2^2 - \frac{1}{|\Omega|} \int_{\Omega} x_1^2 + x_2^2 dx,$$

$\Omega = B_{1,h}(0)$ with 4K triangles and $k = 3$.

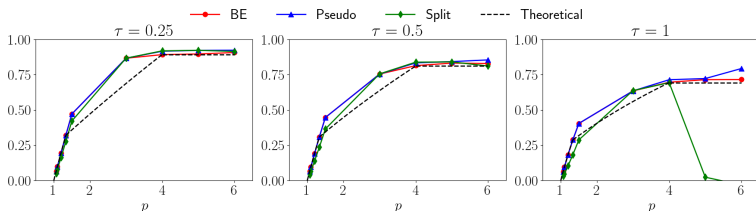


Figure: Comparison of experimental (continuous lines) and theoretical (dashed) convergence rates of BE (red), Pseudo (blue), and Split (green) methods for the p -Stokes problem as a function of p for different values of τ as shown on top of each panel.

Comparing Pseudo method against Newton

As a common stopping criterion, we the residual of the Euler–Lagrange equations of the primal FEM problem

$$\mathcal{R}^*[u_h](v_h) := \int_{\Omega} \mu(|Du_h|^2) Du_h : Dv_h \, dx - F(v_h),$$

and declare convergence when its Euclidean norm $\|\mathcal{R}^*[u_h]\|_{\ell^2} < 1.E-8$ within a maximum of 200 iterations.

p -Laplacian equations

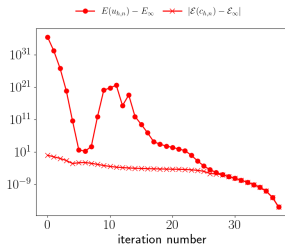
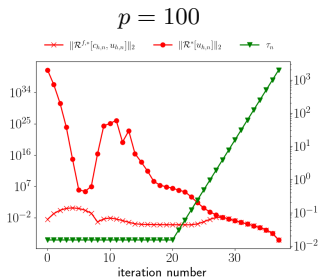
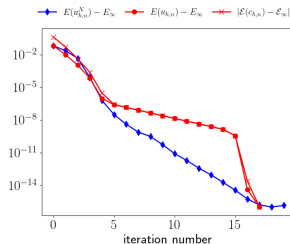
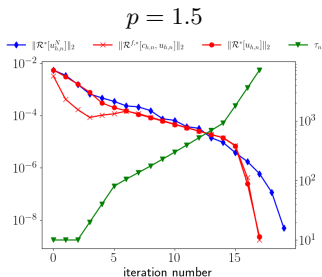
Classical benchmark: L-shaped domain, constant right-hand side²⁶

	$p = 1.5$	$p = 3$	$p = 4$	$p = 8$	$p = 10$	$p = 20$	$p = 40$	$p = 80$	$p = 100$
τ_0	10	9	4	0.5	0.2	0.1	0.05	0.02	0.015
$r = 0$	15 (15)	8 (8)	9 (13)	19 (25)	28 (-)	25 (-)	25 (-)	30 (-)	31 (-)
$r = 1$	18 (16)	8 (8)	10 (15)	18 (-)	28 (-)	25 (-)	28 (-)	33 (-)	34 (-)
$r = 2$	18 (20)	9 (10)	12 (20)	18 (-)	31 (-)	28 (-)	29 (-)	34 (-)	38 (-)

Table: Iteration counts for the Pseudo method compared against Newton (in parentheses) for the power-law model for different model parameters (columns) and refinement levels (rows). The initial time step τ_0 is also reported. The symbol “-” indicates failure to converge in 200 iterations with absolute tolerance $1.E-8$ for $\|\mathcal{R}^*[u_h]\|_{\ell^2}$.

²⁶Balci, Diening, and Storn 2023; Carstensen and Tran 2021.

p -Laplacian equations



Regularized Bingham fluids

Here we consider the Bercovier–Engelman regularization²⁷ of Bingham fluids²⁸

$$\mu(r) = 2\nu + \frac{\sigma}{\sqrt{r + b_\epsilon^2}},$$

where $\sigma > 0$ is the so-called yield stress, $\nu > 0$ the fluid viscosity, and $b_\epsilon > 0$ a regularization parameter. The model satisfies the hypothesis needed for the global existence of the ODE in the concave regime since

$$\mu'(r) = -\frac{\sigma}{2}(r + b_\epsilon^2)^{-3/2} < 0, \quad \forall r > 0.$$

However, $S_\phi^- > 0$ but $S_\phi^+ = 1$.

²⁷Bercovier and Engelman 1980.

²⁸Bingham 1922.

Regularized Bingham fluids

We reproduce the numerical experiment described in²⁹, considering $\Omega = [0, 1]^2$, and analytical solution

$$\mathbf{u}_e = (u_\infty, 0), \quad u_\infty(x) := \begin{cases} 0.25 (0.4^2 - (0.4 - 2x_2)^2), & 0 \leq x_2 \leq 0.2, \\ 0.02, & 0.2 < x_2 < 0.8, \\ 0.25 (0.4^2 - (2x_2 - 1.6)^2), & 0.8 \leq x_2 \leq 1, \end{cases}$$

and set $\sigma = 0.3$ and $\nu = 1$. We use Taylor-Hood pair with $k = 2$, $\tau_0 = 1$.

	$b_\epsilon = 1.e-2$	$b_\epsilon = 1.e-3$	$b_\epsilon = 1.e-4$
$r = 0$	13 (8)	19 (27)	40 (-)
$r = 1$	12 (8)	20 (18)	35 (86)
$r = 2$	12 (7)	19 (22)	37 (92)

Table: Iteration counts for the Pseudo method compared against Newton (in parentheses) for the Bingham model for different model parameters (columns) and refinement levels (rows). The symbol “-” indicates failure to converge in 200 iterations with absolute tolerance $1.E-8$ for $\|\mathcal{R}^*[u_h]\|_{\ell^2}$.

²⁹Heid and Süli 2022a, Section 5.1.

Future work

- Refine ODE convergence theory
- Convergence analysis of fully-discrete algorithms
- Adaptive mesh refinement
- Improve adaptive time-step selection
- Extension to energies depending on ∇u , $\nabla \mathbf{u}$, or $\varepsilon(\mathbf{u})$

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